

QUESTION BANK

Course Code & Name	23IT036 & Security and Privacy in Cloud
Year / Semester	4 / 7
Department	CSE
Regulation	R23

UNIT I: FUNDAMENTALS OF CLOUD SECURITY CONCEPTS

S. No	Questions	CO	BL
	Part - A		
1	What is cloud security?	CO1	K1
2	Define confidentiality.	CO1	K1
3	What is integrity in cloud security?	CO1	K1
4	Define authentication.	CO1	K1
5	What is non-repudiation?	CO1	K1
6	What is access control?	CO1	K1
7	Define cryptography.	CO1	K1
8	What is symmetric key cryptography?	CO1	K1
9	What is public key cryptography?	CO1	K1
10	Define hash function.	CO1	K1
11	Define digital signature and state its purpose.	CO1	K2
12	Differentiate encryption and hashing.	CO1	K2
13	State any two properties of hash functions.	CO1	K2
14	Compare symmetric and asymmetric cryptography.	CO1	K2
15	Explain the purpose of authentication in cloud environments.	CO1	K2
16	Discuss the role of access control in cloud security.	CO1	K2
17	Explain the importance of non-repudiation.	CO1	K2

18	Describe any two applications of cryptography.	CO1	K2
19	Explain the use of digital signatures in cloud transactions.	CO1	K2
20	Explain how hashing is used for password protection.	CO1	K2
Part - B			
1	Explain the various security services used in cloud computing.	CO1	K2
2	Explain authentication protocols used in cloud security.	CO1	K2
3	Explain authentication and digital signatures in cloud security.	CO1	K2
4	Explain the importance of hash functions.	CO1	K2
5	Compare symmetric and asymmetric cryptography.	CO1	K2
6	Explain access control mechanisms in cloud security.	CO1	K2
7	Explain cryptographic techniques used for securing cloud data.	CO1	K2
8	Explain how cryptographic mechanisms ensure confidentiality, integrity, and authentication in cloud computing.	CO1	K2

Unit II: SECURITY DESIGN AND ARCHITECTURE FOR CLOUD

S. No	Questions	CO	BL
Part - A			
1	Define cloud security architecture.	CO2	K1
2	What is secure isolation?	CO2	K1
3	Define virtualization.	CO2	K1
4	What is network segmentation?	CO2	K1
5	Define data retention.	CO2	K1
6	What is data deletion?	CO2	K1
7	Define tokenization.	CO2	K1
8	What is data obfuscation?	CO2	K1
9	What is encryption?	CO2	K1
10	Define Public Key Infrastructure (PKI).	CO2	K1
11	Explain end-to-end access control.	CO2	K2

12	Discuss the need for secure isolation in cloud environments.	CO2	K2
13	Differentiate tokenization and encryption.	CO2	K2
14	Explain inter-tenant network segmentation.	CO2	K2
15	Describe data archiving procedures in cloud computing.	CO2	K2
16	Explain the role of key management in cloud security.	CO2	K2
17	Discuss common attack vectors in cloud environments.	CO2	K2
18	Explain comprehensive data protection in cloud computing.	CO2	K2
19	Explain the use of encryption in protecting cloud data.	CO2	K2
20	Explain a secure isolation strategy used in a multi-tenant cloud.	CO2	K2
Part – B			
1	Explain the security design principles for cloud computing.	CO2	K2
2	Explain comprehensive data protection and end-to-end access control mechanisms in cloud environments.	CO2	K2
3	Explain secure isolation strategies used in multi-tenant cloud infrastructures.	CO2	K2
4	Explain virtualization strategies used to enhance cloud security.	CO2	K2
5	Compare encryption, tokenization and obfuscation techniques used for cloud data protection.	CO2	K2
6	Explain the role of PKI and key management in securing cloud resources.	CO2	K2
7	Explain common attack vectors and threats affecting cloud environments.	CO2	K2
8	Explain the architectural mechanisms used to secure a multi-tenant cloud, including secure isolation, segmentation and data protection strategies.	CO2	K2

Unit – III: ACCESS CONTROL AND IDENTITY MANAGEMENT

S. No	Questions	CO	BL
	Part – A		

1	Define Identity Management.	CO3	K1
2	What is user identification?	CO3	K1
3	Define authentication.	CO3	K1
4	What is authorization?	CO3	K1
5	What is Role-Based Access Control (RBAC)?	CO3	K1
6	Define Multi-Factor Authentication (MFA).	CO3	K1
7	What is Single Sign-On (SSO)?	CO3	K1
8	Define Identity Federation.	CO3	K1
9	Who is an Identity Provider?	CO3	K1
10	What is a Service Consumer?	CO3	K1
11	Differentiate authentication and authorization.	CO3	K2
12	Explain the benefits of Multi-Factor Authentication.	CO3	K2
13	Discuss the purpose of Single Sign-On.	CO3	K2
14	Explain Identity Federation.	CO3	K2
15	What is OS Hardening and Minimization?	CO3	K2
16	Explain verified boot and measured boot.	CO3	K2
17	Describe Intrusion Detection Systems (IDS).	CO3	K2
18	Describe Intrusion Prevention Systems (IPS).	CO3	K2
19	Illustrate RBAC implementation in a cloud environment.	CO3	K3
20	Illustrate Identity Federation using a cloud scenario.	CO3	K3
Part – B			
1	Illustrate the implementation of access control requirements in a cloud environment.	CO3	K3
2	Illustrate the implementation of Role-Based Access Control (RBAC) with a suitable cloud scenario.	CO3	K3
3	Illustrate the implementation of Multi-Factor Authentication (MFA) and Single Sign-On (SSO) in cloud environments.	CO3	K3

4	Demonstrate the working of Identity Federation.	CO3	K3
5	Apply authentication and authorization mechanisms to secure a cloud environment.	CO3	K3
6	Illustrate OS hardening, minimization, and secure boot mechanisms used in cloud systems.	CO3	K3
7	Illustrate the use of IDS and IPS techniques in cloud environments.	CO3	K3
8	Apply Identity and Access Management (IAM) strategies to secure cloud resources.	CO3	K3

Unit - IV: CLOUD SECURITY DESIGN PATTERNS

S. No.	Questions	CO	BL
	Part - A		
1	What is a cloud security design pattern?	CO5	K1
2	Define cloud bursting.	CO5	K1
3	What is geo-tagging?	CO5	K1
4	Define secure cloud interface.	CO5	K1
5	What is cloud resource access control?	CO5	K1
6	Define secure on-premise internet access.	CO5	K1
7	What is secure external cloud access?	CO5	K1
8	State the purpose of cloud bursting.	CO5	K1
9	What is the role of secure cloud interfaces?	CO5	K1
10	What is the purpose of Secure External Cloud architecture?	CO5	K1
11	Explain the importance of geo-tagging in cloud environments.	CO5	K2
12	Discuss the need for secure cloud interfaces.	CO5	K2
13	Discuss the benefits of cloud resource access control.	CO5	K2
14	Describe secure external cloud connectivity.	CO5	K2

15	Explain security concerns in hybrid cloud environments.	CO5	K2
16	Discuss secure on-premise internet access strategies.	CO5	K2
17	Explain the benefits of cloud bursting.	CO5	K2
18	Describe secure cloud integration mechanisms.	CO5	K2
19	Illustrate a cloud bursting scenario with an example.	CO5	K3
20	Apply cloud resource access control in a cloud deployment.	CO5	K3
Part – B			
1	Apply the cloud bursting concept to handle varying workload demands in cloud environments.	CO5	K3
2	Illustrate secure cloud interfaces for protecting cloud resources.	CO5	K3
3	Illustrate cloud resource access control mechanisms with suitable examples.	CO5	K3
4	Analyze the role of geo-tagging in cloud security and resource management.	CO5	K4
5	Compare various cloud security design patterns and their applications.	CO5	K4
6	Illustrate secure on-premise internet access strategies used in hybrid cloud environments.	CO5	K3
7	Analyze the security challenges associated with secure external cloud connectivity.	CO5	K4
8	Design a secure cloud architecture for a healthcare organization using appropriate cloud security design patterns.	CO5	K4

Unit – V: MONITORING, AUDITING AND MANAGEMENT

S. No.	Questions	CO	BL
Part – A			
1	Define proactive activity monitoring.	CO4	K1
2	What is incident response?	CO4	K1
3	Define unauthorized access.	CO4	K1

4	What is malicious traffic?	CO4	K1
5	Define auditing.	CO4	K1
6	What is record generation?	CO4	K1
7	Define an event in cloud monitoring.	CO4	K1
8	What is an alert?	CO4	K1
9	What is a tamper-proof audit log?	CO4	K1
10	Define Security Information and Event Management (SIEM).	CO4	K1
11	Explain the purpose of monitoring unauthorized access.	CO4	K2
12	Discuss the importance of incident response in cloud environments.	CO4	K2
13	Differentiate monitoring and auditing.	CO4	K2
14	Explain the significance of QoS in cloud service management.	CO4	K2
15	Describe user management in cloud environments.	CO6	K2
16	Explain identity management in cloud security.	CO6	K2
17	Discuss the significance of audit logs in cloud environments.	CO4	K2
18	Explain the role of SIEM in cloud security management.	CO4	K2
19	Illustrate the incident response lifecycle with an example.	CO4	K3
20	Apply monitoring techniques to detect cloud security threats.	CO4	K3
Part - B			
1	Illustrate proactive monitoring techniques used in cloud environments.	CO4	K3
2	Apply the incident response lifecycle to a cloud security incident.	CO4	K3
3	Illustrate auditing, record generation, reporting and management mechanisms in cloud platforms.	CO4	K3
4	Illustrate the use of tamper-proof audit logs in cloud auditing.	CO4	K3
5	Illustrate monitoring and auditing mechanisms used in cloud environments.	CO4	K3
6	Illustrate the architecture and functions of Security Information and Event Management (SIEM).	CO4	K3

7	Illustrate privacy-preserving identity management techniques used in cloud environments.	CO6	K3
8	Design a cloud monitoring and incident management framework for detecting and responding to security threats in an enterprise cloud environment.	CO4	K3